

Foundations of Machine Learning

Attendance



March 13th 2026

Session 2

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Session 2 Agenda & Topic

Decision trees & K-NN algorithm

Codebook 2a

Break

Linear regression & practical considerations

Codebook 2b

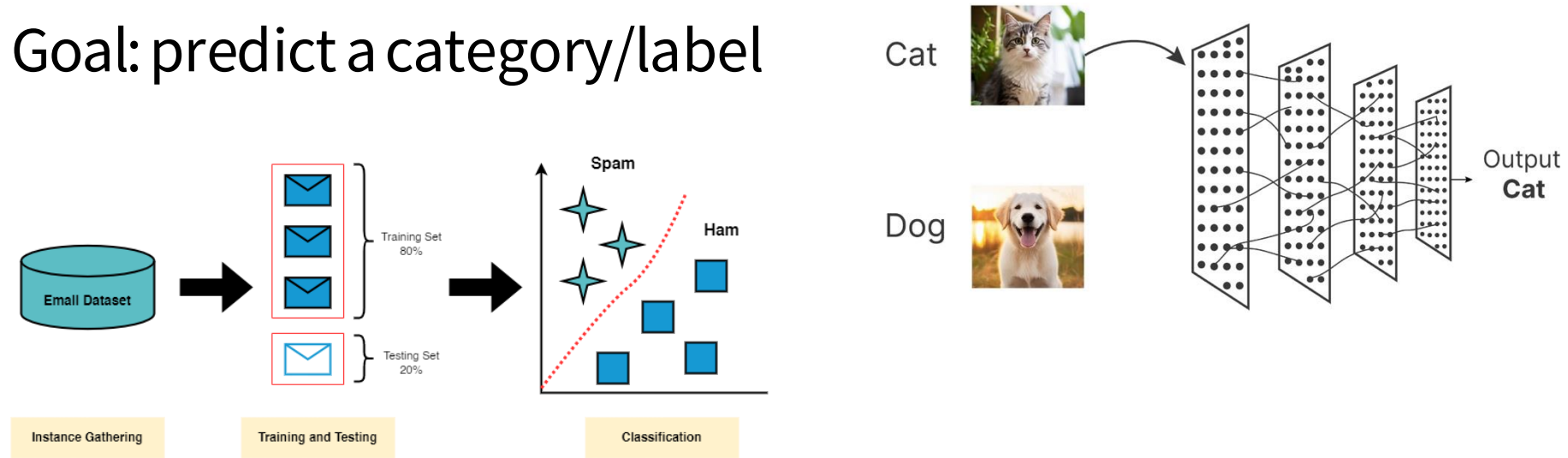
Summary and conclusion



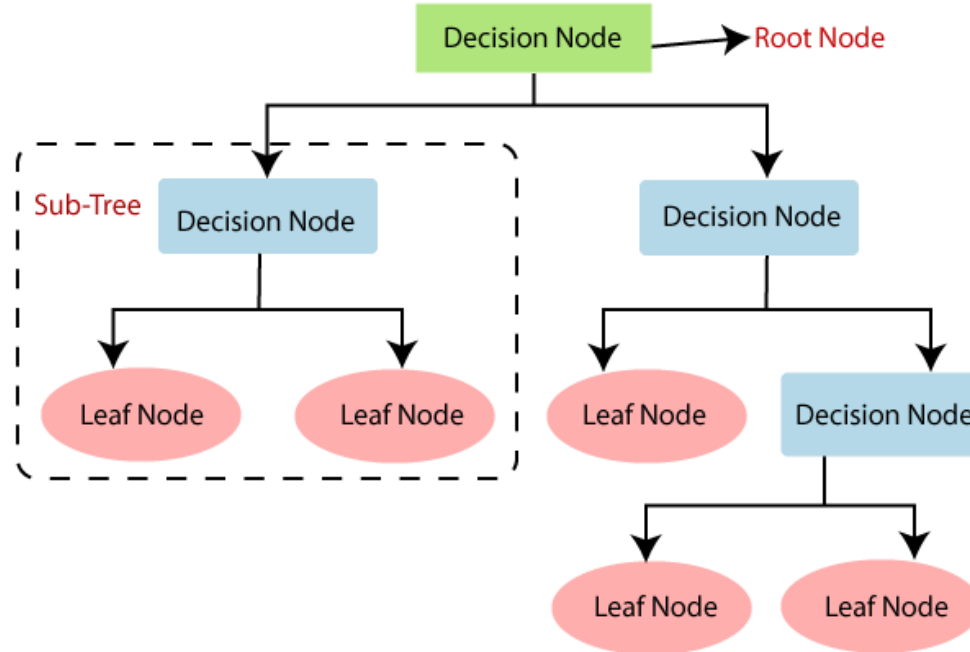
A Short Recap On Classification

A supervised learning task

Goal: predict a category/label



Decision Trees



How does a decision tree make a decision?

1. Picks the best split at each node
2. Keeps splitting until no more gain (of information) from splitting
3. Assigns class based on majority vote in leaf nodes



Decision tree - Example

#	Sex	Class	Age	Survived
1	Female	1st	Adult	 Yes
2	Male	3rd	Adult	 No
3	Female	3rd	Child	 Yes
4	Male	1st	Adult	 No
5	Male	3rd	Child	 Yes
6	Female	2nd	Adult	 Yes

Split by Class

- 1st (rows 1, 4) →  
- 2nd (row 6) → 
- 3rd (rows 2, 3, 5) →   

Split by Sex

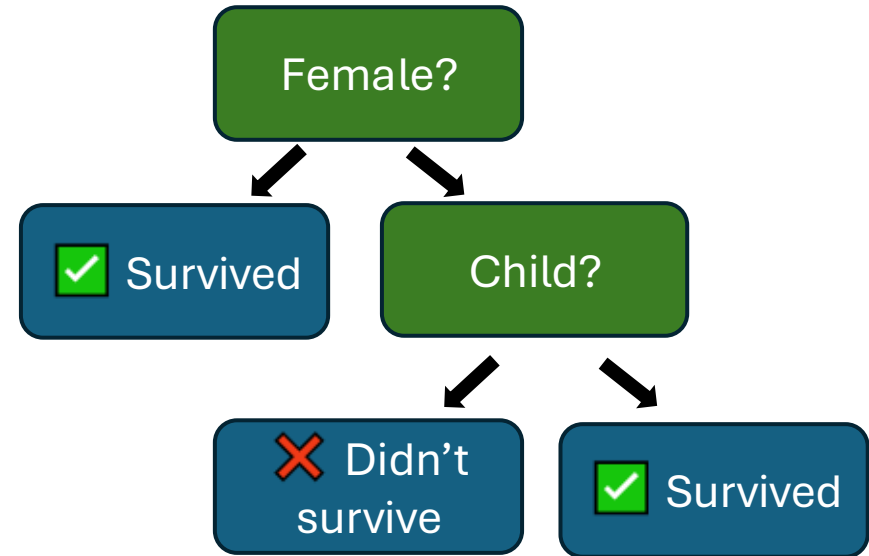
- Female (rows 1, 3, 6) →   
- Male (rows 2, 4, 5) →   

Split by Age

- Adult (rows 1, 2, 4, 6) →    
- Child (rows 3, 5) →  

Decision tree - Example

#	Sex	Class	Age	Survived
1	Female	1st	Adult	✓ Yes
2	Male	3rd	Adult	✗ No
3	Female	3rd	Child	✓ Yes
4	Male	1st	Adult	✗ No
5	Male	3rd	Child	✓ Yes
6	Female	2nd	Adult	✓ Yes
7	Male	1st	Child	??



Decision tree – Pros & Cons

Easy to understand

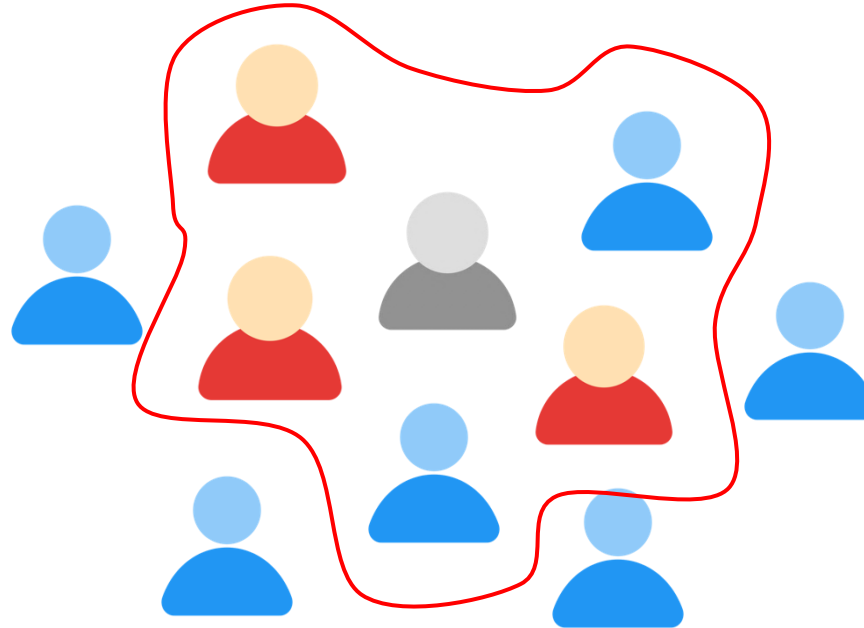
Can handle mixed data types

No need to scale data

Doesn't always generalise well

Tree structure sensitive to small change in data

K-nearest neighbour

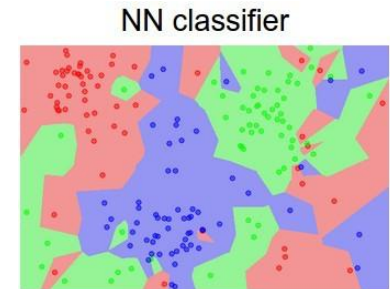
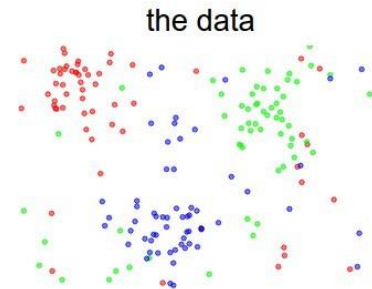


Ask your 5 most similar friends. If most liked it, you probably will too.



How does a K-NN algorithm make a decision?

1. Compute distance between test point and all training points
2. Pick the k closest ones
3. Use majority vote to assign class label



K-NN - Example

	Sex	Age	Budget (£)	Liked?
A	F(0)	27	250	Yes
B	M(1)	30	80	No
C	F(0)	26	220	Yes
D	M(1)	29	200	Yes
E	M(1)	31	90	No
F	M(1)	45	60	No
	F(0)	29	150	??

Step 1: Calculate Distance

Distance = Difference in Sex, Age, Budget

The 5 closest friends: A, C, D, B, E

Step 2: Majority Vote (k = 5)

Among these 5 friends:

• Yes → A, C, D (3 votes)   

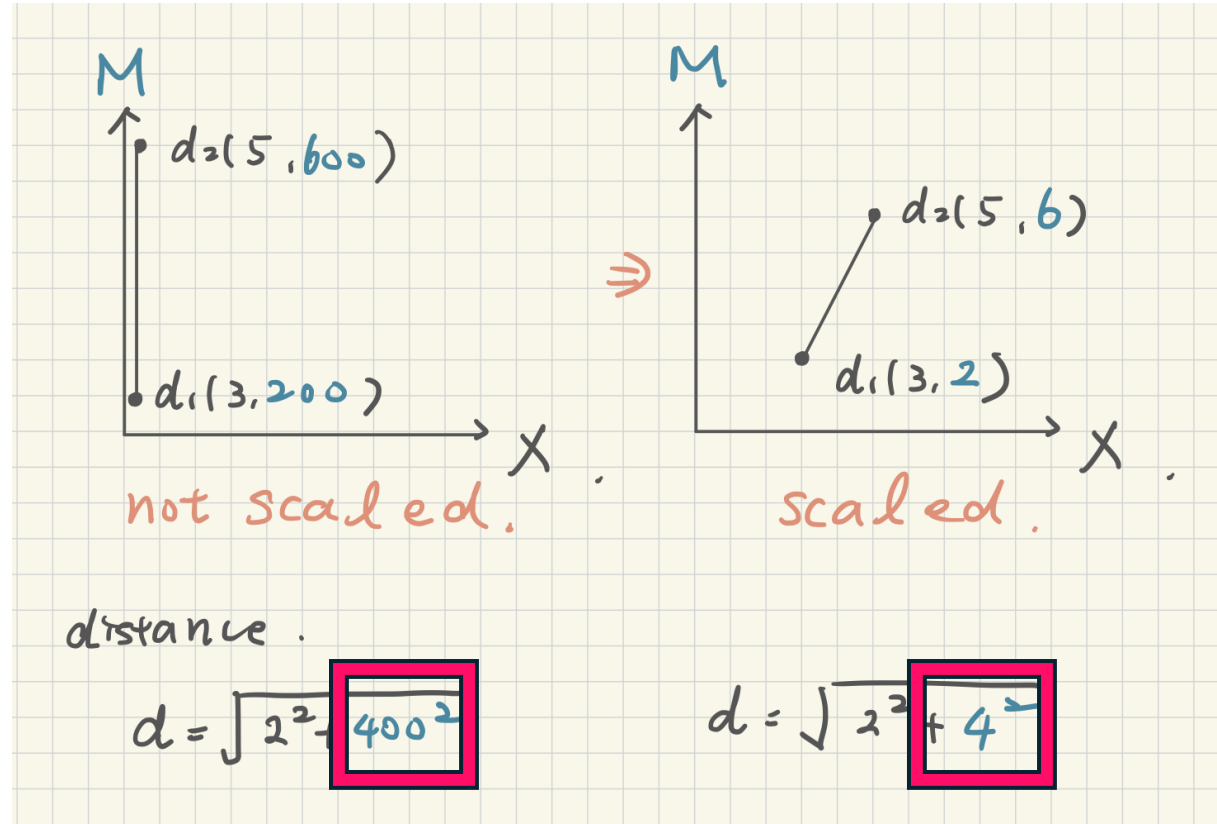
• No → B, E (2 votes)  

Prediction: ?



K-NN – feature scaling

Why does feature scaling matter?



K-NN – Pros & Cons

Easy to understand

No training phase

Works with any number of classes

Slower with large datasets

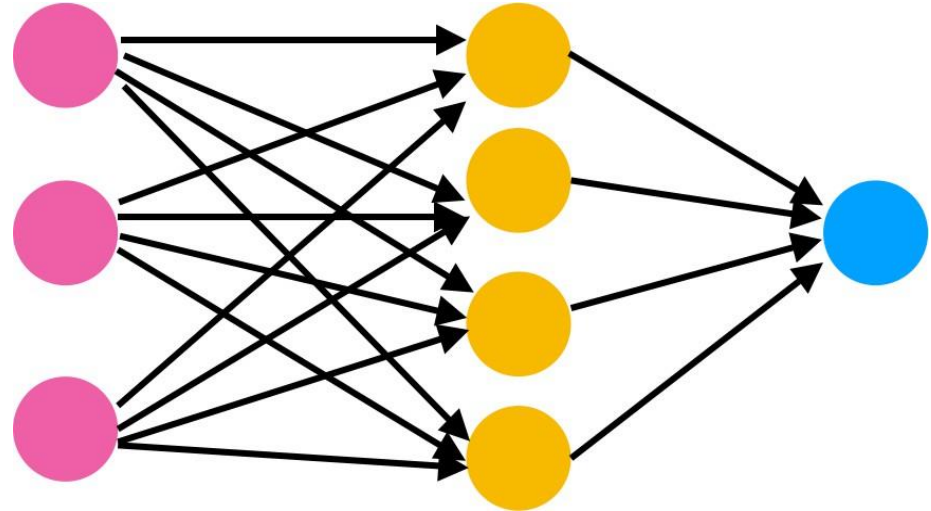
Sensitive to feature scaling

Needs careful choice of 'k'



A short note on neural networks

- Inspired by the brain, Layers of connected “neurons”.
- Great for complex, high-dimensional problems but require more data & computation
- Typically used when simpler models don’t perform well



Method	When?	What to worry about.
Logistic Regression	Quick baseline, linearly separable data	Doesn't capture complex boundaries
Decision Trees	Interpretability, non-linear relationships	Prone to overfitting
K-NN	Simple, low-dimensional problems	Slow with big data, needs scaling
Neural Networks	Large datasets with complex patterns	Requires tuning, hard to interpret





Let's Code

Session 2a

Codebook 2a

Decision Tree

```
``{r}  
# Initialise the Decision Tree Classifier  
# Hint: we need to use y_train ~ .  
  
tree_model <- rpart(y_train ~ ., data = X_train, method = "class",  
control = rpart.control(cp = 0))  
``
```

```
``{r}  
# EXERCISE: Predict using 'tree_model' on 'X_test' and calculate  
accuracy, precision, and recall.  
y_pred_tree <- predict(tree_model, X_test, type = "class")  
``
```

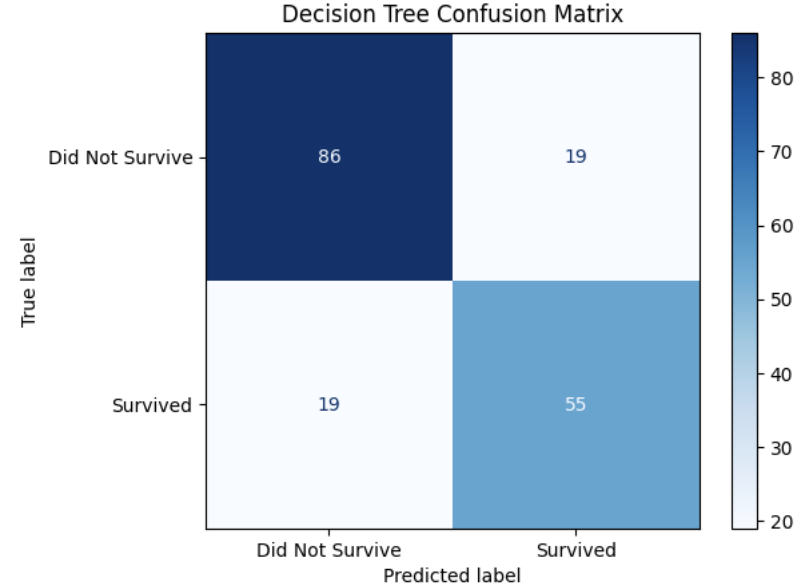
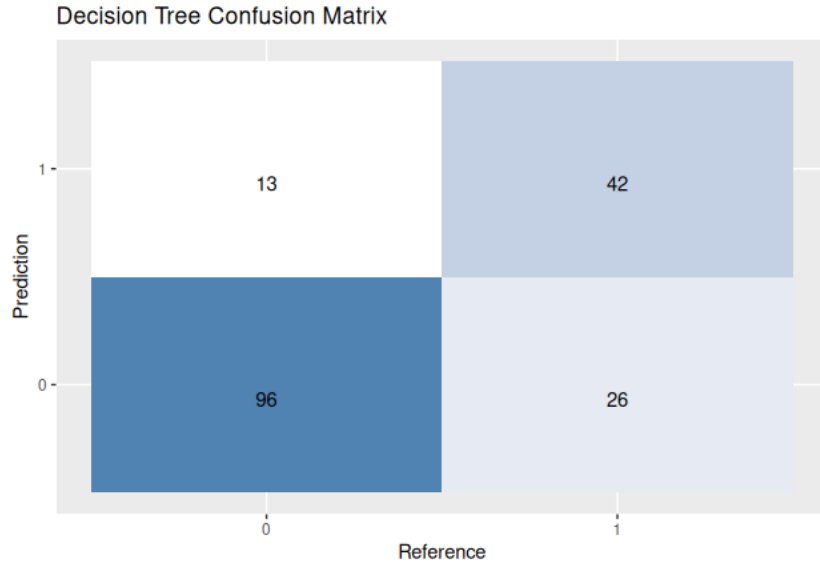
```
1 # Initialise the Decision Tree Classifier  
2 tree_model = DecisionTreeClassifier(random_state=42)
```

```
1 # EXERCISE: Train the Decision Tree model on X_train and y_train.  
2 # Hint: Use .fit() method.  
3  
4 tree_model.fit(X_train, y_train)  
5  
6 # EXERCISE: Predict on X_test and calculate accuracy, precision, and recall.  
7 y_pred_tree = tree_model.predict(X_test)
```

1. Train on X_train, y_train (builds the decision tree)
2. Predict on X_test

Codebook 2a

Decision Tree



Q: Why are the results different between Python and R?



Codebook 2a

Decision Tree

Pruning a decision tree to reduce overfitting and faster training

1. Pre-pruning: stops a decision tree from growing if it won't improve the performance.
2. Post-pruning: remove branches and leaves, if it reduces error.

Criteria: maximum depth, information gain, gini impurity, etc.



Codebook 2a

Decision Tree

Try:

1. Select different sets of features to train the data.
(ex. Remove Sex, and Age), and compare the results.
2. Pre-prune or post-prune the tree

Example for Python

<https://www.kaggle.com/code/arunmohan003/pruning-decision-trees-tutorial>



Codebook 2a

K-NN

```
1 # Initialise the k-NN model with 5 neighbours
2 knn_model = KNeighborsClassifier(n_neighbors=5)

1 # EXERCISE: Train the k-NN model on X_train and y_train.
2 scaler = StandardScaler()
3 X_train_scaled = scaler.fit_transform(X_train)
4 X_test_scaled = scaler.transform(X_test)
5
6 knn_model.fit(X_train_scaled, y_train)
7
8 # EXERCISE: Predict on X_test using the k-NN model, then calculate accuracy
9 y_pred_knn = knn_model.predict(X_test_scaled)
```

```
# EXERCISE: Train and use the k-NN model on X_train and y_train. Use k=5.
# Ensure all inputs are scaled numerically for k-NN
X_train_scaled <- scale(X_train)
X_test_scaled <- scale(X_test, center = attr(X_train_scaled, "scaled:center"),
scale = attr(X_train_scaled, "scaled:scale"))

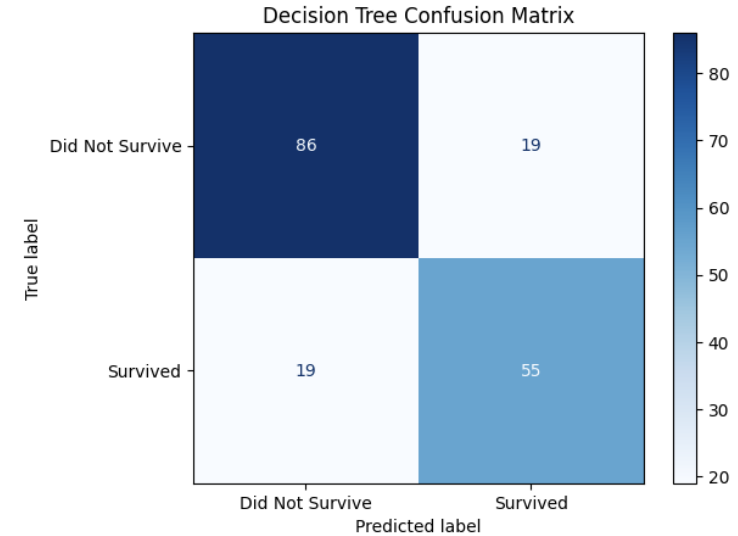
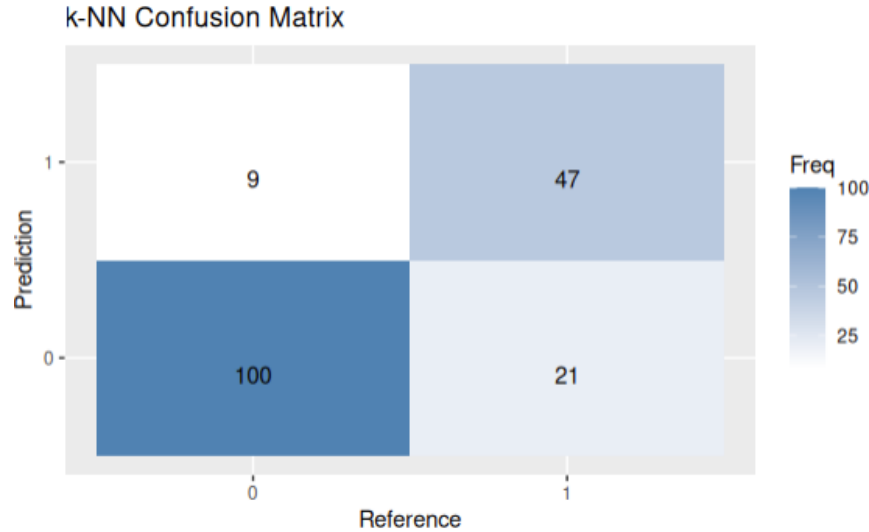
y_pred_knn <- knn(train = X_train_scaled, test = X_test_scaled, cl = y_train, k =
5)
```

1. Scale `X_train`, `X_test`
2. Train on `X_train_scaled`, `y_train`
3. Predict on `X_test_scaled`



Codebook 2a

K-NN



Codebook 2a

K-NN

Try:

1. Compare the results before and after scaling.
2. Compare the result of using different K



Codebook 2a

Compare decision tree and K-NN

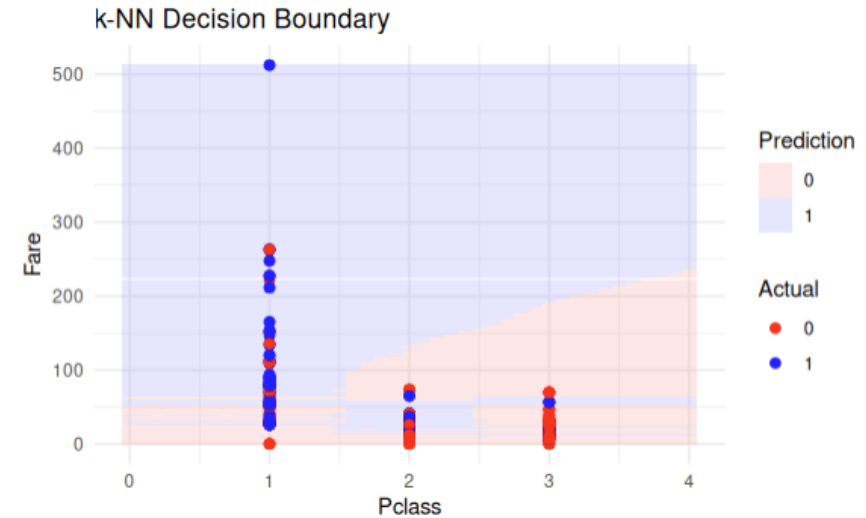
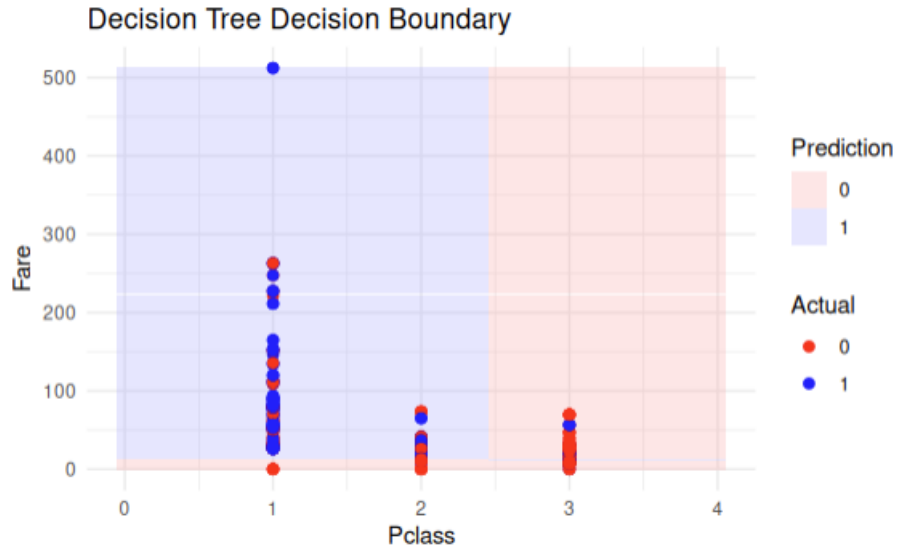
Model <chr>	Accuracy <dbl>	Precision <dbl>	Recall <dbl>
Decision Tree	0.7796610	0.7636364	0.6176471
k-NN	0.8305085	0.8392857	0.6911765



Codebook 2a

Compare decision tree and K-NN

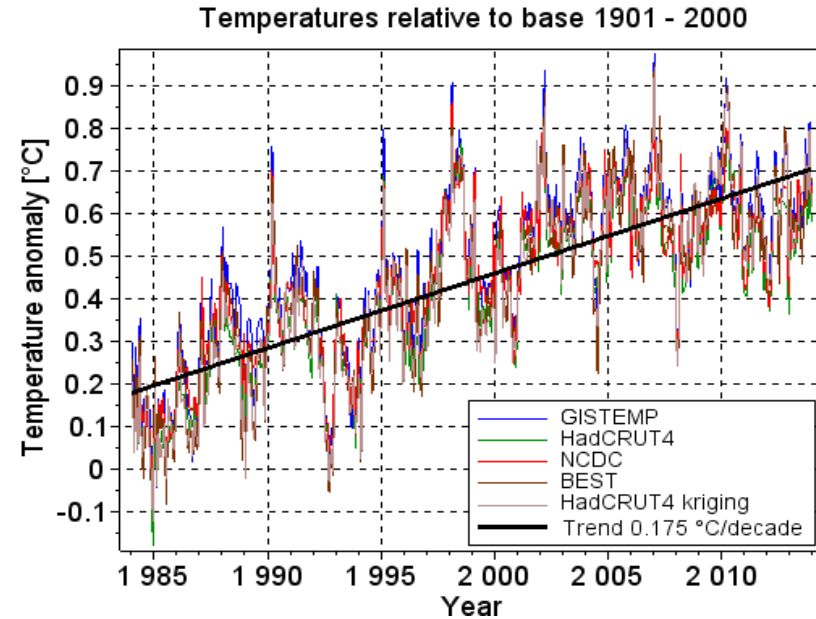
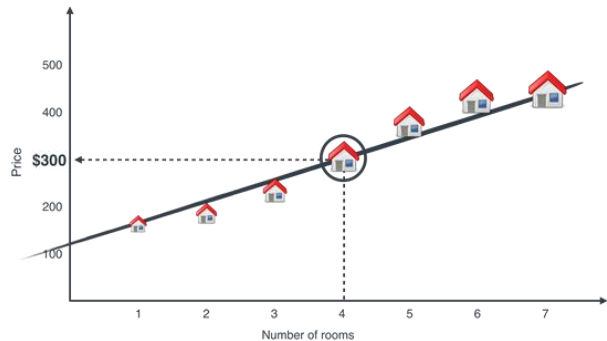
Model <chr>	Accuracy <dbl>	Precision <dbl>	Recall <dbl>
Decision Tree	0.7796610	0.7636364	0.6176471
k-NN	0.8305085	0.8392857	0.6911765



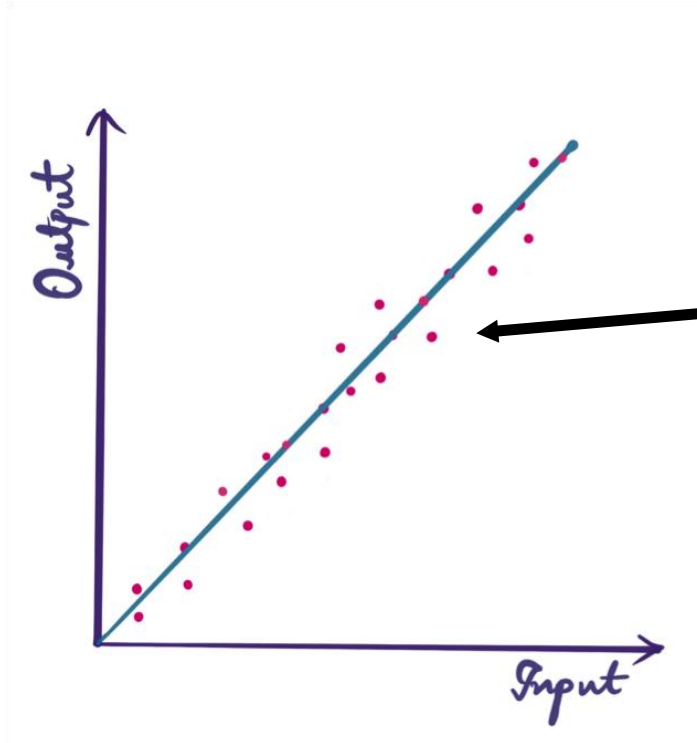
Regression

A supervised learning task

Goal: Predicting continuous outcomes



How does regression work



Linear regression

What is linear regression in machine learning

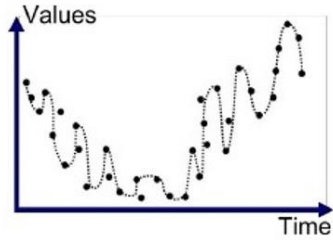
Traditional (modelling-as-analysis)	Machine Learning Approach
Fit the model once	Train/test split (or cross-validation)
Interpret coefficients	Prioritise predictive performance
Check p-values, confidence intervals	Track metrics like MAE, MSE, R^2
Choose features based on theory	Use feature selection, regularisation
Assume model is correct	Evaluate on unseen data
Focus on statistical assumptions	Focus on robustness and generalisation



How do we know if a regressor did well?

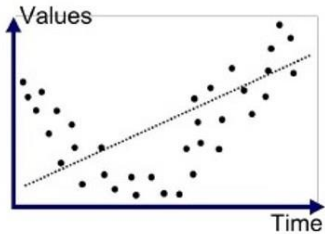
Metric	What is it?	Formula	Good value?
MAE Mean Absolute Value	Average absolute difference	$\frac{1}{n} \sum_1^n y_i - \hat{y}_i $	Low
MSE Mean Square Value	Square large error	$\frac{1}{n} \sum_1^n (y_i - \hat{y}_i)^2$	Low
R^2	Proportion of variance	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Close to 1

Practical Considerations



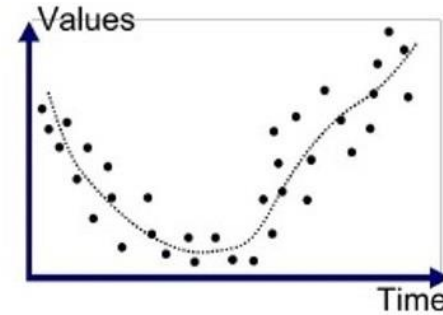
Overfitted

Model performs well on training data but poorly on new data



Underfitted

Model is too simple to capture patterns

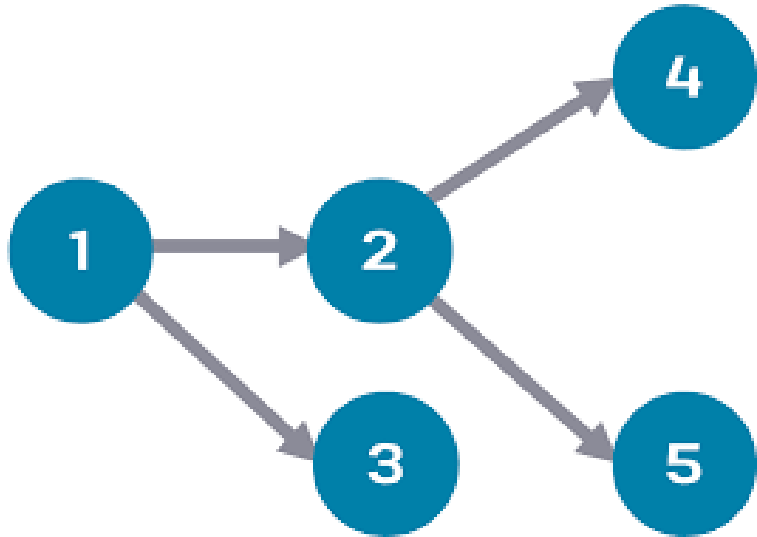


Good Fit/Robust

Fix with simpler/more complex models, regularisation, or **more data**

Practical Considerations

Feature selection

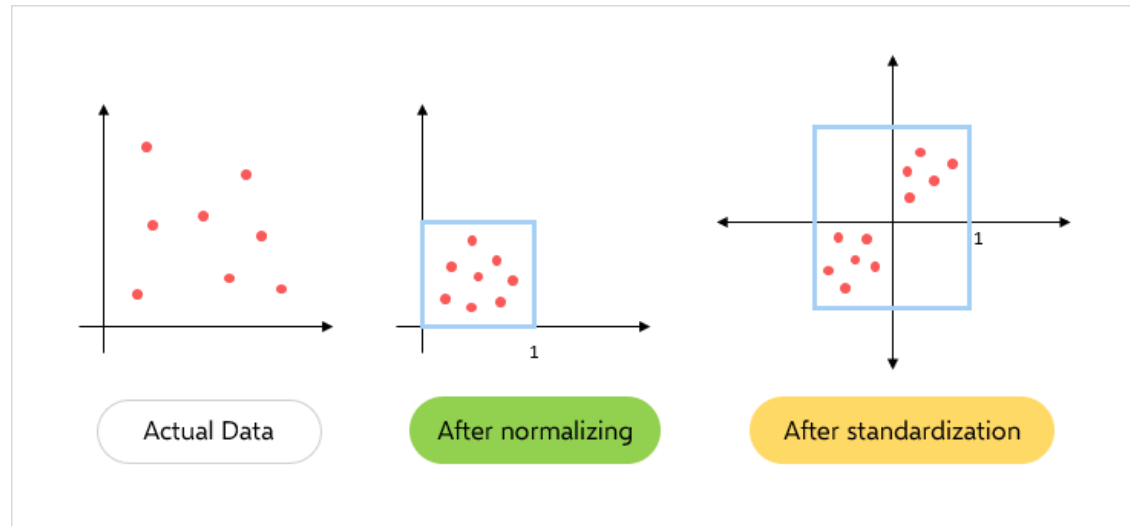


Some features may be noisy or irrelevant.

Use causal inference techniques to establish a directed acyclic graph first when considering features.

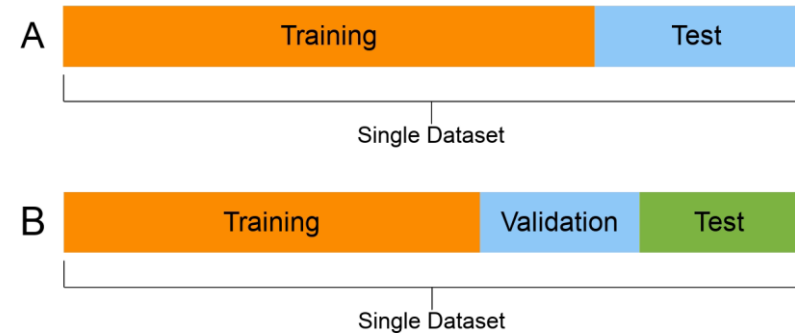
Practical Considerations

Scaling...some models (e.g. k-NN, linear regression) need scaled inputs



Practical Considerations

Data leakage...when information from outside the training set is used improperly.





Let's Code

Session 2b

Codebook 2a

Linear regression

```
```{r}  
Initialise the Linear Regression model
EXERCISE: Train the Linear Regression model on X_train and y_train.
linear_reg <- lm(y_train_white ~ ., data = X_train_white)
```
```

```
```{r}  
EXERCISE: Predict the quality scores on X_test using the trained model.
y_pred <- predict(linear_reg, newdata = X_test_white)
```
```

```
# Initialise the Linear Regression model  
linear_reg = LinearRegression()
```

```
# EXERCISE: Train the Linear Regression model on X_train and y_train.  
linear_reg.fit(X_train_white, y_train_white)
```

```
# EXERCISE: Predict the quality scores on X_test using the trained model.  
y_pred = linear_reg.predict(X_test_white)
```

1. Train on X_train_white, y_train_white
2. Predict on X_test_white

Codebook 2a

Linear regression

| term
<chr> | estimate
<dbl> | std.error
<dbl> | statistic
<dbl> | p.value
<dbl> |
|---------------|-------------------|--------------------|--------------------|------------------|
| (Intercept) | -18.9837340 | 6.84349626 | -2.773982 | 5.563737e-03 |
| alcohol | 0.3464156 | 0.01637366 | 21.156890 | 3.711676e-94 |
| sulphates | 0.4934442 | 0.11341344 | 4.350844 | 1.390736e-05 |
| `citric acid` | 0.1155199 | 0.10613266 | 1.088448 | 2.764643e-01 |
| density | 20.3928705 | 6.75794921 | 3.017612 | 2.564153e-03 |
| pH | 0.2093362 | 0.08713195 | 2.402520 | 1.632886e-02 |

What variables have the most predictive power?



Codebook 2a

Linear regression

Can we generalise the model?

```
# EXERCISE: Predict the quality of red wine.  
  
y_pred_red = linear_reg.predict(X_red)
```

```
# EXERCISE: Predict the quality of red wine using the model.  
y_pred_red <- predict(linear_reg, newdata = X_red)
```

| Dataset
<chr> | Mean_Squared_Error
<dbl> | Mean_Absolute_Error
<dbl> | R_squared
<dbl> |
|---------------------|-----------------------------|------------------------------|--------------------|
| White Wine Test Set | 0.6599146 | 0.6344821 | 0.19793498 |
| Red Wine (New Data) | 0.6188692 | 0.6350923 | 0.05046533 |



Session 2 Wrap-up



Additional Resources

- Introduction to Machine Learning with Python: A Guide for Data Scientists by Sebastian Raschka and Vahid Mirjalili
- Machine Learning with R by Brett Lantz
- Artificial Unintelligence: How Computers Misunderstand the World by Meredith Broussard
- TheAIMirror: How to Reclaim Out Humanity in an Age of Machine Thinking by Shannon Ballor



Feedback form



<https://forms.office.com/r/YYNrquvNr8>

